DEMO SCRIPT

FETCH: Inventory Management System with Predictive Analytics for Risha Pet Supplies

System presentation and live demonstration

Presenters	Juan Carlo Reusora (Speaker 1) and Joylinda Jugan (Speaker 2)
Length	About 25 minutes, then questions
System	https://risha-pets.onrender.com
Roles shown	Administrator, Supplier and Cashier
How to read the script	DO = what to click on screen. SAY = the words to say (in your own natural voice; it does not have to be word for word). POINT AT = what the panel should notice.

The speakers can swap any part. Practise the demo at least twice from the "Before the demo" checklist, with a stopwatch.

Run sheet

#	Part	Speaker	Minutes	Clock
1	Opening	Juan Carlo	1	0:00
2	What FETCH is	Juan Carlo	2	0:01
3	The problem it solves	Juan Carlo	3	0:03
4	Business requirements	Joylinda	3	0:06
5	Live demo A — the administrator prepares stock and orders	Joylinda	5	0:09
6	Live demo B — the supplier confirms and ships	Juan Carlo	2	0:14
7	Live demo C — the administrator receives; supplier rating	Joylinda	2	0:16
8	Live demo D — the cashier sells	Juan Carlo	3	0:18
9	Live demo E — closing the day; reports and the forecast	Joylinda	4	0:21
10	Closing	Juan Carlo	1	0:25

If time is short, skip the starred (★) beats. They are useful but not needed to show the requirements are met.

Before the demo

FETCH is live and uses the shop's real records. Everything done in the demo is saved for real, so prepare it this way:

Accounts and windows

- Open three separate browser windows, because one browser window keeps only one person signed in: Chrome for the Administrator, a Chrome Incognito window for the Cashier, and Safari (or another browser) for the Supplier.
- Sign in to all three before the panel arrives, then leave each on its home page (Dashboard, Point of Sale, Supplier Portal).
- Use a demo supplier whose email is your own (for example reusorajuan+demo@gmail.com) and link the Supplier account to it. The reorder in the demo emails that supplier, so no real supplier receives a demo order.
- Have the cashier's starting cash already saved for today, so the demo starts at the till.

Things to prepare in the data

- A demo product of that demo supplier, at or below its reorder level, so Reorder lists it (for example "DEMO Dog Treats", stock 5, reorder level 10).
- Optional for the expiry beat (★): a demo product whose expiration date has already passed.
- Know which product you will sell at the till, and its price.

Things the demo changes for real — and how to undo them

Demo action	What it leaves behind
The sale at the till	Part 9 voids it from the cashier's request, so its items return to stock and it leaves the totals.
Cash taken out	Stays in today's cash movements. Use a small amount and the reason "Demo".
End of day (cashier and whole shop)	Closes today's drawer: no more sales today until an administrator clicks Reopen. Do these steps only if the shop is closed for the day; otherwise explain them without saving.
The demo product and purchase order	Delete the demo product after the defense if you do not want it in the records.

Backup

- Check the internet and open https://risha-pets.onrender.com 30 minutes before; the first visit after a quiet period can take up to a minute to wake up.
- Keep screenshots of each part on the laptop in case the internet fails, and say so calmly: "We will continue with screenshots of the same steps."

1. Opening

1.1 SPEAKER 1 — Juan Carlo	
DO	Title slide, or the FETCH sign-in page on the projector.
SAY	"Good morning. We are Juan Carlo Reusora and Joylinda Jugan, and we will present FETCH: an Inventory Management System with Predictive Analytics for Risha Pet Supplies." "We will explain what FETCH is, the problem it solves for the shop, and the business requirements it was built to meet. Then we will show the whole system working, the way the shop would use it in one day."

2. What FETCH is

2.1 SPEAKER 1 — Juan Carlo	
SAY	"Risha Pet Supplies is a neighbourhood pet supply shop. It sells pet food, treats, grooming products, supplements and accessories, and many of these products have expiration dates." "FETCH is the shop's web-based system. It keeps the inventory, runs the point of sale at the counter, handles purchase orders with suppliers, and forecasts how much of each product will sell in the next 30 days, so it can tell the owner what to order and by when." "It is used by three roles. The Administrator runs the back office. The Cashier sells at the counter. And the Supplier has its own portal to confirm and ship the shop's orders."

2.2 SPEAKER 1 — Juan Carlo	
SAY	"FETCH runs in a web browser on a laptop or a phone, so there is nothing to install. The pages are built with HTML, CSS and JavaScript, the server with Node.js and Express, the data is kept in a MySQL database, and the forecast is made in Python with scikit-learn."

3. The problem it solves

Taken from Chapter 1 of the paper (Introduction, Significance of the Study and Objectives of the Study).

3.1 SPEAKER 1 — Juan Carlo	
SAY	"The pet care industry keeps growing, because people now treat their pets as part of the family and buy them good food, grooming and health products." "For a small shop like Risha Pet Supplies, that means a long list of products to keep track of: food, grooming supplies, toys, accessories and health products, many of them with expiration dates."

3.2 SPEAKER 1 — Juan Carlo	
SAY	"Small pet stores usually manage all of this by hand. They count what is on the shelves, write the numbers in spreadsheets, and decide when and how much to order from memory and feeling." "This causes clear problems. Popular items sell out, which means lost sales and unhappy customers. Other products sit on the shelves, which wastes money and space. And poor ordering decisions hurt the relationship with suppliers." "A small business cannot afford these mistakes, because it works with little money to spare."

3.3 SPEAKER 1 — Juan Carlo	
SAY	"In short, FETCH answers five problems: stock records that do not match the shelf, overstocking, stockouts, errors from manual processing, and poor coordination with suppliers." "So our objective was to develop FETCH: an inventory system that forecasts demand with an ensemble of Random Forest and Gradient Boosting, uses barcode scanning, reorders automatically when stock reaches the reorder point, tracks stock and expiration dates in real time, and works with the suppliers. We then tested it with unit testing, system testing and user acceptance testing, and evaluated its usability with the CSUQ."

4. Business requirements

4.1 SPEAKER 2 — Joylinda	
DO	Show the requirements table below (as a slide).
SAY	"These are the business requirements we agreed with the owner of Risha Pet Supplies and signed. There are seven groups. During the demo we will point out where each one is met."

From the signed Business Requirements form (FETCH_Business Requirements, 3 pages).

BR	Business requirement (signed)	Shown in the demo
1	The owner (Admin) manages the system: 1.1 an administrator account with full access; 1.2 staff accounts with limited POS access; 1.3 user access control and permissions.	Part 5 (the administrator); Part 8 (the cashier is kept on the Point of Sale)
2	Admin and Staff manage inventory and sales: 2.1 add, edit and track products and stock; 2.2 view the forecasts and reorder recommendations; 2.3 POS with barcode scanning; 2.4 stock updated after each sale; 2.5 no duplicate entries.	Part 5 (products; the SKU and barcode cannot be duplicated); Part 8 (selling); Part 9 (forecast)
3	Complete inventory records: 3.1 product information (SKU, name, brand, category, species); 3.2 stock levels and reorder points; 3.3 expiration dates and batch/lot numbers; 3.4 supplier purchase order history; 3.5 sales history with date stamps.	Parts 5, 7 and 9
4	Predictive analytics and automation: 4.1 forecasting with Random Forest + Gradient Boosting; 4.2 reorder points from past sales; 4.3 seasonal trends; 4.4 purchase orders when stock reaches the reorder point; 4.5 suppliers notified automatically.	Part 5 (Reorder emails the supplier); Part 9 (Stock planning)
5	Reports and alerts: 5.1 daily, weekly and monthly reports; 5.2 low-stock alerts; 5.3 alerts for expiry within 30 days, reorder point, stockout and overstock; 5.4 dashboards with charts; 5.5 export.	Part 5 (Dashboard); Part 9 (Reports)
6	Data quality and business rules: 6.1 expired products cannot be sold; 6.2 every entry validated (format, range, consistency); 6.3 data integrity; 6.4 audit trail of changes.	Part 5 (the product form); Part 8 (the till); Part 9 (Activity log)
7	Technical and integration: 7.1 barcode scanner; 7.2 thermal receipt printer; 7.3 automated daily backups; 7.4 MySQL database.	Part 8 (scan and receipt); backups run daily by themselves

Added during development

- The Supplier Portal, where suppliers confirm, ship and date the shop's orders (part of the supplier management in Objective 1).
- A star rating for each supplier, from its recorded deliveries.
- Cash control at the counter: starting cash, cash in and out, and the end-of-day count.
- Void requests: a cashier asks, the administrator decides.

5. Live demo A — the administrator prepares the shop

5.1 SPEAKER 2 — Joylinda	
DO	In the Administrator window, show the Dashboard.
SAY	"This is the Administrator's Dashboard, the first page after signing in. At a glance the owner sees the total products, total revenue, how many products need restocking, how many are expiring soon, and the value of the stock on hand." "Below are the recent sales, which refresh by themselves, the products to restock, the products expiring in the next 30, 60 and 90 days, and the purchase orders."
POINT AT	*Restock Needed and Expiring Soon, and the charts — BR 5.2 to 5.4.* *Inventory value is counted at the selling price.*

5.2 SPEAKER 2 — Joylinda	
DO	Click Inventory in the sidebar. Click the Low Stock chip, then the Expired chip, then All.
SAY	"Inventory lists every product with its stock and status. The chips filter the list: Low Stock shows only products at or below their reorder level, and Expired only those past their date. The count on each chip matches the list."

5.3 SPEAKER 2 — Joylinda	
DO	Click Add. Click Save product with the form empty.
SAY	"To add a product, the form asks only for what the shop needs. If we save it empty, each required field says exactly what is missing, and the cursor goes to the first one. This is BR 6.2: every entry is validated, so no wrong or missing data is saved."
POINT AT	*"Enter the product name." "Choose a category from the list." "Choose the product type. The SKU is made from it." "Enter the unit price (the selling price)." "Choose a supplier. A product needs one so it can be reordered."*

5.4 SPEAKER 2 — Joylinda	
DO	Type a product name and choose a category. Choose a Product type, for example Dry food.
SAY	"The SKU is not typed. We choose the product type, and the system makes the SKU from it with the shop's running number — here it shows the next one. Once saved, the SKU never changes, and two products can never share one. This is BR 2.5 and 3.1."
POINT AT	*The SKU box fills in by itself, for example DRY-121.*

5.5 SPEAKER 2 — Joylinda	
DO	Type 120.555 as the Unit price and leave the field. Then type 150 as the Cost price.

SAY	"Every field checks its value as we leave it. A price with three decimal places is refused, and a cost higher than the selling price is caught, because the shop would lose money on each sale."
POINT AT	*"Use at most 2 decimal places, for example 120.50."* *"The cost price (₱150.00) is higher than the unit price …"*

5.6 SPEAKER 2 — Joylinda	
DO	Correct the prices (for example 120.50 and 90), enter the stock and reorder level, choose the supplier, then click Save product.
SAY	"With correct values the product is saved, and the confirmation shows the SKU it was given. This is BR 2.1 and 3."
POINT AT	*"Product added … with the SKU …"*

5.7 ★ SPEAKER 2 — Joylinda	
DO	Click Stock on the expired demo product. Choose "Add stock (received, found, returned)" and enter a quantity.
SAY	"Stock changes are recorded with a reason. For a product that has already expired, the system asks for the new batch's expiration date, and the expired units are removed and replaced, not added on top. This is BR 3.3 and 6.1."
POINT AT	*"The (n) expired will be removed. Stock will go from (n) to (m)."*

5.8 SPEAKER 2 — Joylinda	
DO	Click Reorder. Tick the demo product, click Reorder Selected, then Yes, Create Orders.
SAY	"Reorder lists the products at or below their reorder level. With one confirmation, FETCH creates the purchase orders and emails each one to its supplier. This is BR 4.4 and 4.5."
POINT AT	*"Purchase orders sent"*

6. Live demo B — the supplier confirms and ships

6.1 SPEAKER 1 — Juan Carlo	
DO	Switch to the Supplier window (Safari). Refresh the Supplier Portal.
SAY	"Now we are the supplier. The supplier has its own portal and sees only its own orders and products — never the shop's sales or other suppliers. This is the supplier management in our first objective, and BR 1.3." "The order the shop just sent is here, waiting for confirmation."

6.2 SPEAKER 1 — Juan Carlo	
DO	Click Confirm order, leave the date empty and click Confirm order. Then choose a delivery date and click Confirm order.
SAY	"The supplier confirms the delivery date. The same validation applies here: without a date it says what is needed."
POINT AT	*"Choose the date you will deliver"* *"Order confirmed."*

6.3 SPEAKER 1 — Juan Carlo	
DO	Click Mark as shipped, enter a receipt number such as DR-1001, then click Mark as shipped.
SAY	"When the goods leave, the supplier marks the order shipped with its delivery receipt number. The shop sees this immediately, without a phone call."
POINT AT	*"Order marked as shipped."*

7. Live demo C — receiving, and the supplier's rating

7.1 SPEAKER 2 — Joylinda	
DO	Switch to the Administrator window. Click Suppliers, then the Orders view. Click Receive into stock on the order, enter the new batch's expiration date, then confirm.
SAY	"Back at the shop, the order shows as on its way. When it arrives, one click receives it: the quantities are added to stock and the expiration date of the new batch is recorded."

7.2 SPEAKER 2 — Joylinda	
DO	Look at the Rating column, then click the supplier's stars.
SAY	"Every supplier is rated with stars, worked out only from what was recorded about its deliveries in the last 12 months. It gets one star for each condition it meets: at least one order received; at least 75 percent on time; at least 90 percent on time; at least 90 percent delivered in full; and an average delivery of 7 days or less." "A supplier meets the shop's standard at 4 stars or more. The supplier sees the same rating on its own portal."
POINT AT	*Each condition is ticked or crossed, with its figure.*

8. Live demo D — the cashier sells

8.1 SPEAKER 1 — Juan Carlo	
DO	Switch to the Cashier window (Incognito) on the Point of Sale.
SAY	"This is the Point of Sale, the only page the cashier uses. The cashier's name and the date are at the top. If the cashier tries to open another page, FETCH brings them back here. This is BR 1.2 and 1.3." "Expired and sold-out products are not among the product cards; they sit under "Can't be sold", so they cannot be sold by mistake. This is BR 6.1."

8.1b ★ SPEAKER 1 — Juan Carlo	
DO	If the barcode scanner is connected: press F2 and scan a product.
SAY	"Products can be added by scanning their barcode, which is faster and avoids picking the wrong item. This is BR 2.3 and 7.1."

8.2 SPEAKER 1 — Juan Carlo	
DO	Click Charge with the cart empty. Click a product card. Enter less cash than the total and click Charge.
SAY	"The till checks every step as well. An empty cart cannot be charged, and if the customer gives less cash than the total, the sale does not go through."
POINT AT	*"Cart is empty"* *"Insufficient cash — customer gave less than the total"*

8.3 SPEAKER 1 — Juan Carlo	
DO	Enter enough cash, click Charge, then Complete sale.
SAY	"With enough cash, the sale is completed: the change is shown, the receipt prints on the thermal printer, and the stock goes down at once. This is BR 2.3, 2.4 and 7.2."
POINT AT	*"Sale completed! Change: ₱…"*

8.4 ★ SPEAKER 1 — Juan Carlo	
DO	Click Cash in / out, choose Taken out, enter a small amount and the reason "Demo", click Record cash taken out, then Record it.
SAY	"Any cash that leaves the drawer for a reason other than a sale is recorded with who took it and why, so the count at the end of the day still adds up."
POINT AT	*"Recorded"*

8.5 SPEAKER 1 — Juan Carlo

DO	Click My sales today, click Ask for void on the demo sale, enter a reason such as "Wrong item rung up", then click Send request.
SAY	"If the cashier makes a mistake, they cannot delete or void a sale themselves. They ask, with a reason, and the administrator decides."
POINT AT	*"Sent to a manager"*

8.6 SPEAKER 1 — Juan Carlo	
DO	Click End of day (only if the shop is closed for the day; otherwise just open it and point at the expected cash).
SAY	"At closing, the cashier counts the drawer. FETCH already knows what should be there — the starting cash, plus cash sales, plus cash put in, minus cash taken out — and shows if the drawer is balanced, over or short."

9. Live demo E — closing the day, reports and the forecast

9.1 SPEAKER 2 — Joylinda	
DO	Switch to the Administrator window. Click Reports, then the Void requests tab. Click Void sale on the request, then confirm.
SAY	"The cashier's request is here with the reason. The administrator voids the sale: its items return to stock and it is left out of every total, but it stays on record marked Voided, and the Activity log keeps who did it and when. This is BR 6.4."
POINT AT	*"Sale voided and its items returned to stock"*

9.2 ★ SPEAKER 2 — Joylinda	
DO	Click Sales, then End of day. Show the expected cash (save the count only if the shop is closed).
SAY	"The administrator also closes the day for the whole shop, adding up every drawer."

9.3 SPEAKER 2 — Joylinda	
DO	Click Reports. Click This month, then Custom with two dates and Apply. Click Export CSV.
SAY	"Reports show revenue, sales, the average sale and items sold for any period, by day, week or month, with the best sellers and sales by category. Any report can be exported. This is BR 5.1 and 5.5."

9.4 SPEAKER 2 — Joylinda	
DO	Click the Stock planning tab.
SAY	"This is the predictive part of FETCH. For every product it forecasts how many will sell in each of the next 30 days, using two machine-learning models, Random Forest and Gradient Boosting, trained on this shop's own sales and averaged together. Each reads the last 120 days of a product's sales." "A product that is too new for the models gets a simple trend, and a product with less than 14 days of sales gets a cautious first estimate, and the page says so. This is BR 4.1 and 2.2."

9.5 SPEAKER 2 — Joylinda	
DO	Point at What to reorder, then Too much stock.
SAY	"The forecast is turned into an instruction. What to reorder lists each product to order, with the reason, the stock on hand, what is already on order, the date to order by, and how many to order. It allows for the supplier's delivery time and keeps safety stock, so the shelf does not run empty while waiting." "Too much stock shows where money is tied up, and Likely to expire before it sells warns before an expiry loss happens. This is BR 4.2."

9.5b SPEAKER 2 — Joylinda	
DO	Scroll to Busiest days of the week and Sales by month.
SAY	"These charts show the shop's patterns over the week and over the year, the seasonal trends the forecast learns from. This is BR 4.3."

9.6 SPEAKER 2 — Joylinda	
DO	Scroll to How good is the forecast.
SAY	"Finally, FETCH is honest about its own accuracy. It tests the forecast on recent weeks it did not train on and shows how close it came, so the owner knows how far to trust it."

10. Closing

10.1 SPEAKER 1 — Juan Carlo	
DO	Return to the Dashboard, or the closing slide.
SAY	"To sum up: FETCH keeps the shop's stock right with every sale and delivery, checks every input, warns before stock runs out or expires, connects the shop with its suppliers and rates them, controls the cash at the counter, and turns the shop's own sales history into a clear reorder plan with its accuracy shown." "It meets the seven groups of business requirements we signed with the owner, for the three roles that run the shop: the administrator, the cashier and the supplier." "Thank you. We are ready for your questions."

If asked about limits

- From the signed form: one location only; no e-commerce, online payment, accounting or payroll, CRM or loyalty program.
- The forecast needs 3 to 12 months of sales history, learns only from Risha Pet Supplies' own past sales, and cannot predict events that never happened before. Products with little history get a cautious estimate, and the page says so.
- The counter takes cash only.

Likely questions and answers

Question	Answer
Why Random Forest and Gradient Boosting?	Both learn patterns in the shop's sales such as the day of the week, recent averages and trends, and they handle the ups and downs of daily sales well. They make different kinds of mistakes, so averaging the two is steadier than either one alone.
How do you know the forecast is accurate?	FETCH tests it on recent weeks the models did not learn from and compares the forecast with what was really sold. The result is shown on the Stock planning page under "How good is the forecast", and today's forecast is corrected by half of the miss in that test, category by category.
When and how much does it say to order?	When: once the stock falls to the reorder point, which is the sales expected during the supplier's delivery time (7 days) plus a safety stock of 1.65 × the forecast's spread × √7, enough for about 95% of days (never below the product's own reorder level). How much: the forecast for the next 30 days plus the 7 days of delivery time, less what is in stock.
What happens to a product that is new?	With less than 14 days of sales it gets a cautious first estimate; from 14 to 41 days a simple trend over its last 8 weeks; from 42 days the two models. The page says which one is used.
How is the supplier rating worked out?	Only from recorded deliveries in the last 12 months: one star each for at least one order received, at least 75% on time, at least 90% on time, at least 90% delivered in full, and an average delivery of 7 days or less. 4 stars or more meets the standard.
Can expired products be sold?	No. They are not among the till's product cards, scanning one shows that it has expired, and the server refuses the sale even if the screen is bypassed.
What happens to expired stock when new stock arrives?	The new batch's expiration date must be entered, and the expired units are written off as spoiled and replaced, not added on top.
How is the system secured?	Everyone signs in. Each role sees only its own pages and records. Passwords must have at least 6 characters with an uppercase letter, a lowercase letter, a number and a special character, and five wrong passwords in 15 minutes lock the sign-in for a while. New accounts confirm their email with a 6-digit code, and changes are kept in the Activity log.
What if two products get the same SKU?	They cannot. The SKU is made by the system from the product type and the next running number, a number is never given twice even after a product is deleted, and two products saved at the same moment still get different numbers.
Why is the inventory value higher than what the shop paid?	It is counted at the selling price, so it shows what the stock would bring in if it all sold. Stock planning shows money tied up at cost.
How did you evaluate usability?	With the CSUQ (Computer System Usability Questionnaire): system usefulness, information quality, interface quality and overall satisfaction. [Give your results here.]

Question	Answer
Chapter 1 names Java Spring Boot for the backend, but the system runs on Node.js. Why?	[Agree on your answer before the defense.]
The signed limitations say local operation and Windows only, but FETCH is online and runs on a Mac. Why?	[Agree on your answer before the defense.]
Who evaluated the system?	Chosen by purposive sampling: one owner and one staff member of Risha Pet Supplies as the main users, 5 to 10 suppliers for the reordering and notifications, 5 to 8 technical experts for the design, functions and forecast accuracy, and 5 to 10 owners or managers of other small pet shops for its relevance, through interviews, system testing, UAT and the CSUQ.
How was the system tested?	With a Unit Test Plan (each module, five checks: completeness, format, range, consistency and database), a System Test Plan following the screens in order, and a User Acceptance Test for each of the three roles, plus automated tests of the server.